SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program/s: B.Tech — Year: II Semester: IV

Stream/s: Artificial Intelligence

Subject: Machine Learning — Time: 03 hrs (10:00am to 1:00pm)

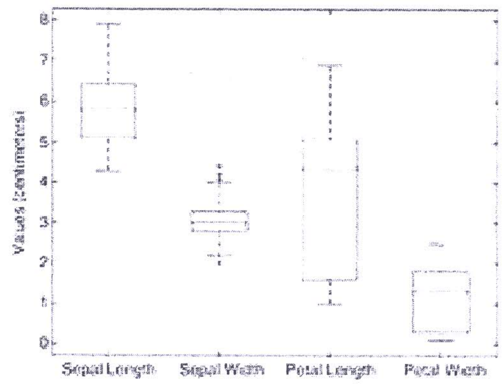
Date: 28 / 06 / 2023 — No. of Pages: 6

Marks: 100

Re-Examination (2021-22/2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		**Answer briefly: (Each question carries 4 marks)**	[20]
CO-1; SO-2; BL-2	a.	Explain the data science life cycle with the help of schematics. Mention the most important challenges in building a machine learning model.	
CO- 1; SO-2; BL-3	b.	For the following vectors, x and y, calculate the indicated similarity or distance measures. a. x = (1, 1, 1, 1), y = (2, 2, 2, 2) cosine, correlation, Euclidean b. x = (0, 1, 0, 1), y = (1, 0, 1, 0) cosine, correlation, Euclidean, Jaccard c. x = (0, −1, 0, 1), y = (1, 0, −1, 0) cosine, correlation, Euclidean d. x = (1, 1, 0, 1, 0, 1), y = (1, 1, 1, 0, 0, 1) cosine, correlation, Jaccard e. x = (2, −1, 0, 2, 0, −3), y = (−1, 1, −1, 0, 0, −1) cosine, correlation	
CO- 1; SO-2; BL-4	c.	Describe how a box plot can give information about whether the value of an attribute is symmetrically distributed. What can you say about the symmetry of the distributions of the attributes shown in Figure? 	

CO-2; SO-2; BL-2	d.	For Linear regression, please derive an equation for all thetas in form of matrix representation for each iteration using gradient descent.	
CO-3; SO-2; BL-2	e.	What is the use of a validation set in classification models in machine learning? How can you solve issues of missing data in a data set for all kind of data types.	

Q2

CO-3; SO-3; BL-3

[20]

a. Predict the class label for a test sample (A = 0, B = 1, C = 0) using the naive Bayes approach.

b. Predict the class label for a test sample (A = 0, B = 1, C = 0) using the naive Bayes approach but with Laplace estimate.

c. Compare the output from (a) and (b), which method is better and why?

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

d. For the one-dimensional data shown below, Classify the data point x = 4.6 according to its 1-, 3-, 5-, and 9-nearest neighbors (using majority vote)

X	0.5	3.0	4.5	4.6	4.9	5.2	5.3	5.5	7.0	9.5
Y	-	-	+	+	+	-	-	+	-	-

Q3 CO-3; SO-3; BL-3			[20]

a. Using Apriori algorithm for the below data, Find
   - The frequent itemsets.
   - Association rules for frequent itemsets with the largest number of items.

Assume that minimum support (s =2) and minimum confident threshold (c = 60%)

TID	Items
T1	I1, I2, I5
T2	I2, I4
T3	I2, I3
T4	I1, I2, I4
T5	I1, I3
T6	I2, I3
T7	I1, I3
T8	I1, I2, I3, I5
T9	I1, I2, I3

b. You are asked to evaluate the performance of two classification models, M1 and M2. The test set you have chosen contains 26 binary attributes, labeled as A through Z. The below table shows the posterior probabilities obtained by applying the models to the test set. (Only the posterior probabilities for the positive class are shown). As this is a two-class problem,
$P(-)=1 - P(+)$ and $P(-|A, \ldots, Z)=1 - P(+|A, \ldots, Z)$.
Assume that we are mostly interested in detecting instances from the positive class:
   - Plot the ROC curve for both M1 and M2. (You should plot them on the same graph.) Which model do you think is better? Explain your reasons.
   - For model M1, suppose you choose the cutoff threshold to be t = 0.5. In other words, any test instances whose posterior probability is greater than t will be classified as a positive example. Compute the precision, recall, and F-measure for the model at this threshold value.
   - Repeat the analysis using the same cutoff threshold on model M2. Compare the F-measure results for both models. Which model is better? Are the results consistent with what you expect from the ROC curve?

Instance	True Class	P(+|A,….., Z, M1)	P(+|A,….., Z, M2)
1	+	0.73	0.61
2	+	0.69	0.03
3	-	0.44	0.68
4	-	0.55	0.31
5	+	0.67	0.45
6	+	0.47	0.09
7	-	0.08	0.38
8	-	0.15	0.05
9	+	0.45	0.01
10	-	0.35	0.04

Q4

CO-2; SO-2;
BL-3

[20]

a. What is Lasso and Ridge regression? Which of these are used for feature reduction (Explain with schematics?)
b. What is Ensemble learning? What is the difference between Bagging and Boosting techniques?
c. For the below Data set build a XGBoost regressor, use learning rate=0.5 and lambda=1, gamma=0. Build the M0 and M1 learners only.

Years of experience	**Gap**	**Annual Salary (in 100k)**
1	N	4
1.5	Y	4
2.5	Y	5.5
3	N	7
5	N	7.5
6	N	9

Q5

CO-3; SO-3;
BL-3

[20]

What are Kernels in support vector machines (give one example of a kernel), Why are they used?
b- For the below data set build a one split regression tree. Draw the one-split tree.
c- For the second split use a CV=25% for 1st split as a split stopping criterion, draw the 2-split tree.
d- What will be the output prediction from the regression tree if you had an input of (Outlook=Rain, Temp.=High, Humidity=High, Wind=Strong)

Day	**Outlook**	**Temp.**	**Humidity**	**Wind**	**Golf Players**
1	Sunny	Hot	High	Weak	25
2	Sunny	Hot	High	Strong	30
3	Overcast	Hot	High	Weak	46
4	Rain	Mild	High	Weak	45
5	Rain	Cool	Normal	Weak	52
6	Rain	Cool	Normal	Strong	23
7	Overcast	Cool	Normal	Strong	43
8	Sunny	Mild	High	Weak	35
9	Sunny	Cool	Normal	Weak	38
10	Rain	Mild	Normal	Weak	46
11	Sunny	Mild	Normal	Strong	48
12	Overcast	Mild	High	Strong	52
13	Overcast	Hot	Normal	Weak	44
14	Rain	Mild	High	Strong	30

Q6 CO-3; SO-2; BL-3		a. Mention the difference between generative and discriminative classifiers, using schematics and give an example for each. b. Define what is High Bias and High Variance in classifiers. How can this be detected in our model design? c. Consider the training examples shown in below table for a binary classification problem.	[20]

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

i. Compute the Gini index for the overall collection of training examples.
ii. Compute the Gini index for the Customer ID attribute.
iii. Compute the Gini index for the Gender attribute.
iv. Compute the Gini index for the Car Type attribute using multiway split.
v. Explain why Customer ID should not be used as the attribute test condition even though it has the lowest Gini.

Q7 CO-3; SO-2; BL-3		a. Please explain with schematics the difference between hierarchical and partitional clustering, mention an algorithm used for each. b. For the below table, please perform **complete linkage agglomerative** clustering and plot the dendrogram, showing the clusters for cluster size=3.	[20]

Student_ID	Marks
1	10
2	7
3	28
4	20
5	35

c. Measure the cluster cohesion (SSE) and cluster separation (SSB) for number of clusters=3, also get the SSE and SSB if we only have one cluster.

d. For the below 2 dimensional feature space, please use DBSCAN to identify the core/border/noise points (use Euclidean distance as the distance measure).

Form clusters with ε=3.5 and MinPts=3

S1	5	7
S2	8	4
S3	3	3
S4	4	4
S5	3	7
S6	6	7
S7	6	1
S8	5	5